



PowerShell Conference Europe

Next Up:

Jess Pomfret & Rob Sewell

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PowerShell Conference Europe

AI close to your data

Jess Pomfret & Rob Sewell

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Jess Pomfret She/Her

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Microsoft Data Platform MVP, dbatools contributor, hardcore data nerd



Rob Sewell He/Him

Zure (UK)

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Your speakers after riding up Cheddar Gorge

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SQL Server 2025

The AI-ready enterprise database from ground to cloud

Best-in-class security
and performance

AI built-In

Made for
developers

Cloud agility
through Azure

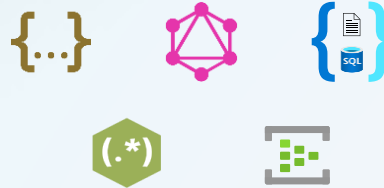
aka.ms/sqlserver2025

SQL Server 2025

The AI-ready enterprise database from ground to cloud



AI built-in



Develop modern
data applications



Integrate your data
with Fabric



Secure by default



Mission critical engine



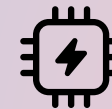
Connected
with Arc



Assisted by
Copilots



Benchmark
leader



Optimized for
latest hardware

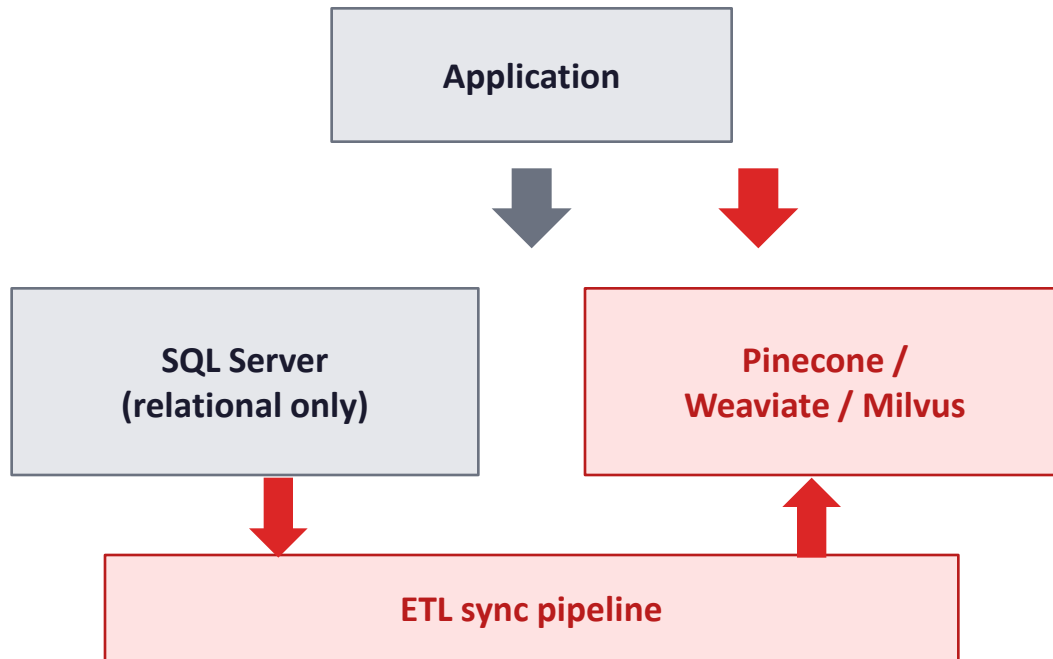


Built for all
platforms

Pattern 1: Where vectors live

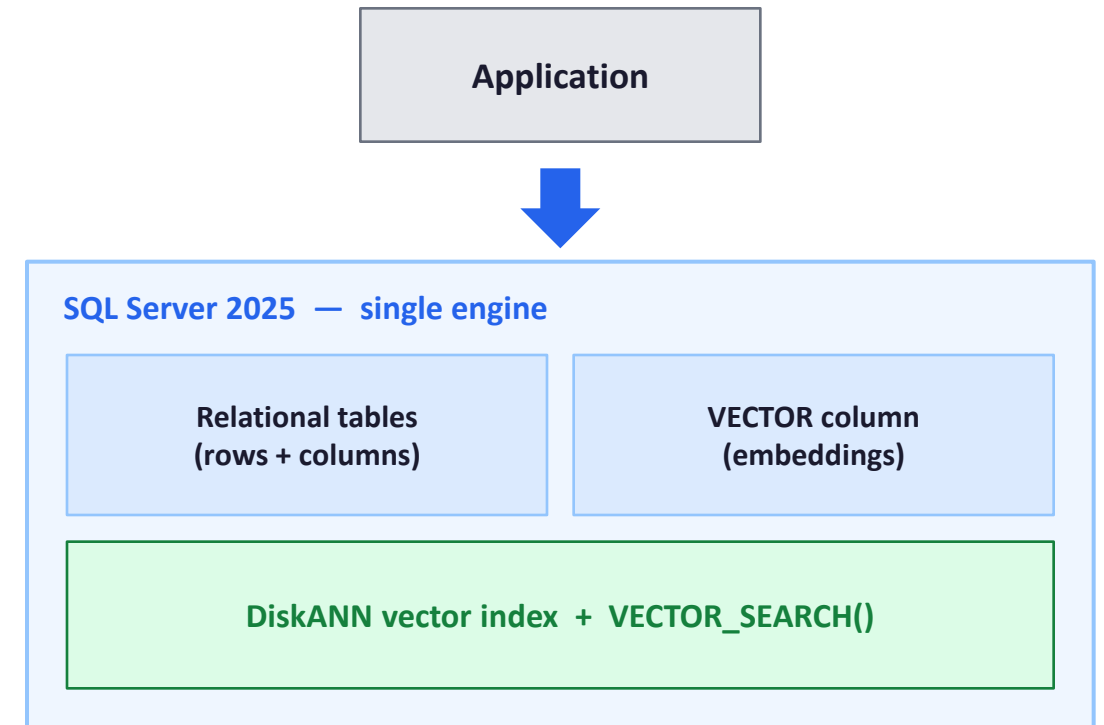
Vector storage architecture for semantic search and RAG

✗ Bolt-on vector database



- Two stores to keep in sync
- Separate security model + auth
- Data drift, dual-write failures

✓ Vectors inside SQL 2025

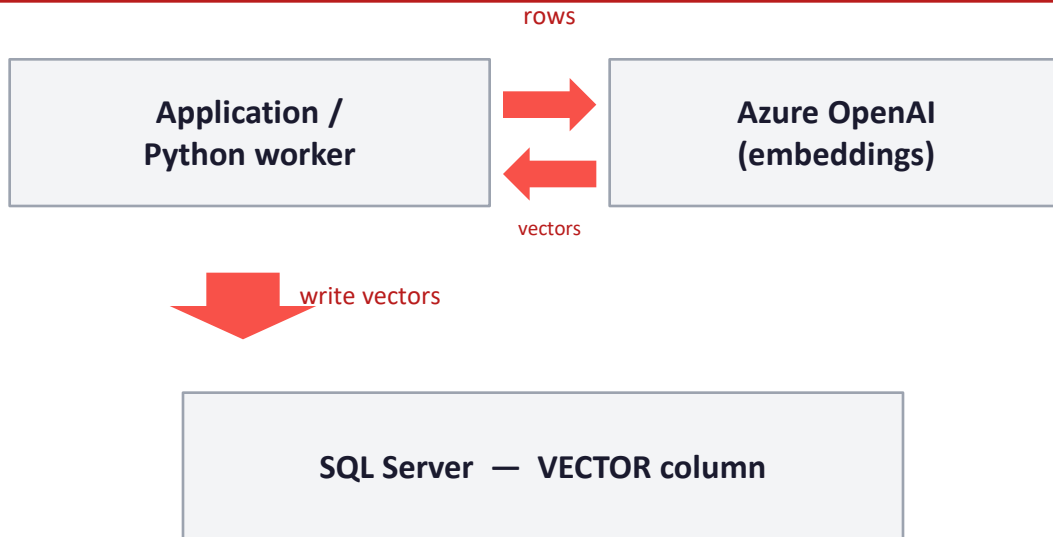


- One store, one backup, one security boundary
- Join vectors with relational data in one query
- No ETL, no drift

Pattern 2: Where embeddings are generated

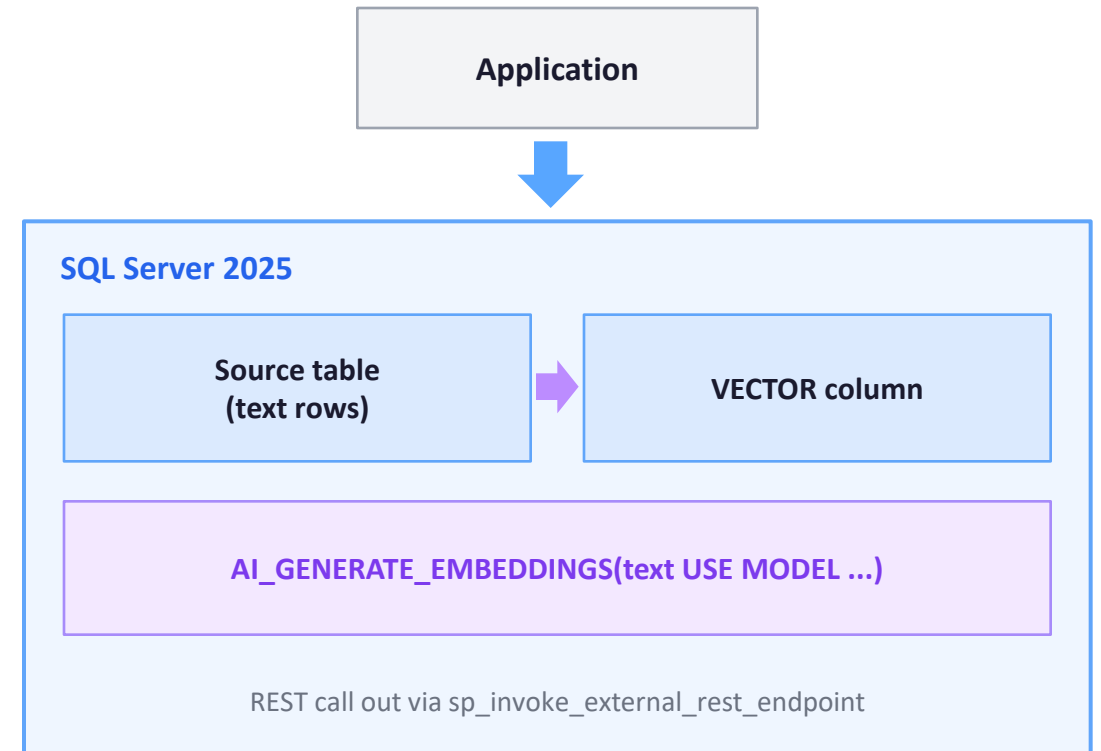
Turning rows into vectors at ingest and at query time

✗ App-side embedding generation



- Data leaves DB; extra network hops
- App owns model config + retries
- Embeddings drift from rows on update
- Hard to embed at query time

✓ AI_GENERATE_EMBEDDINGS in-engine

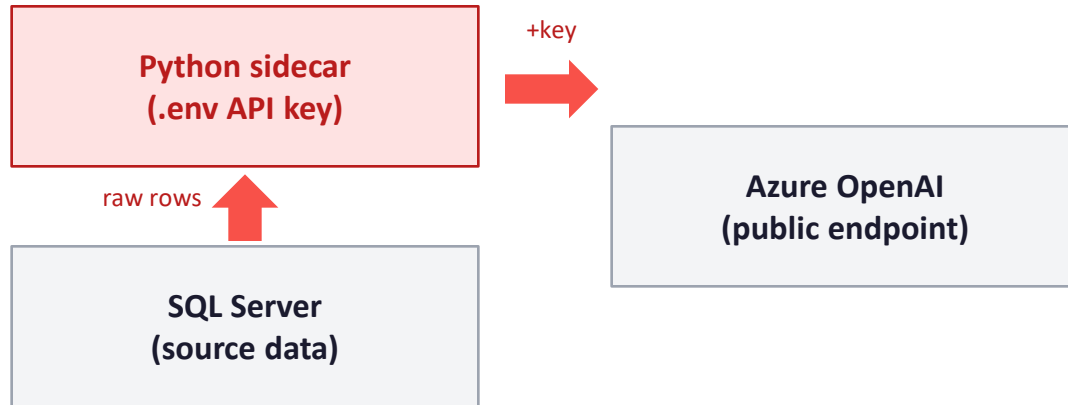


- ✓ Embeddings generated inside SQL, joined in one query
- ✓ Refresh embeddings via trigger; no drift

Pattern 3: Credentials & data egress

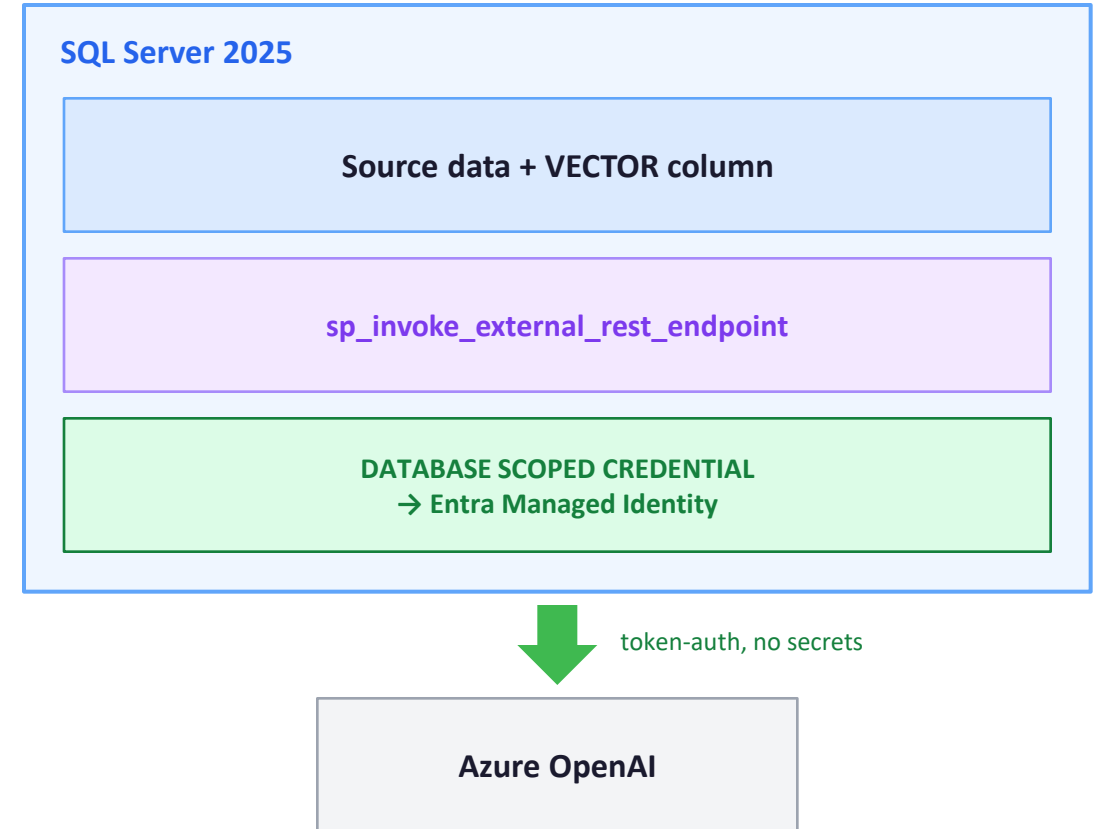
How SQL Server reaches the embedding model — and what data crosses the boundary

✗ Hardcoded keys, raw egress



- Long-lived API key in config / .env
- Raw source rows leave DB boundary
- No managed identity, no rotation
- Audit trail lives outside the database

✓ Managed identity + REST in-engine

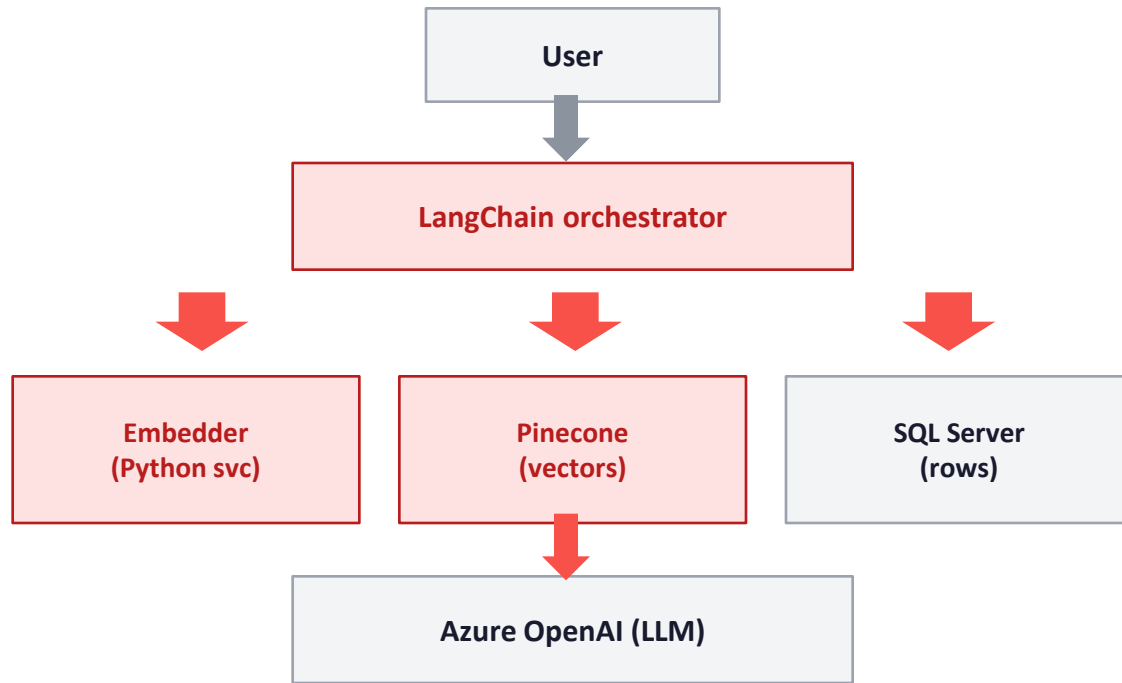


- ✓ No secrets in config; rotation handled by Entra
- ✓ Source rows never leave SQL boundary unencrypted

Pattern 4: End-to-end RAG architecture

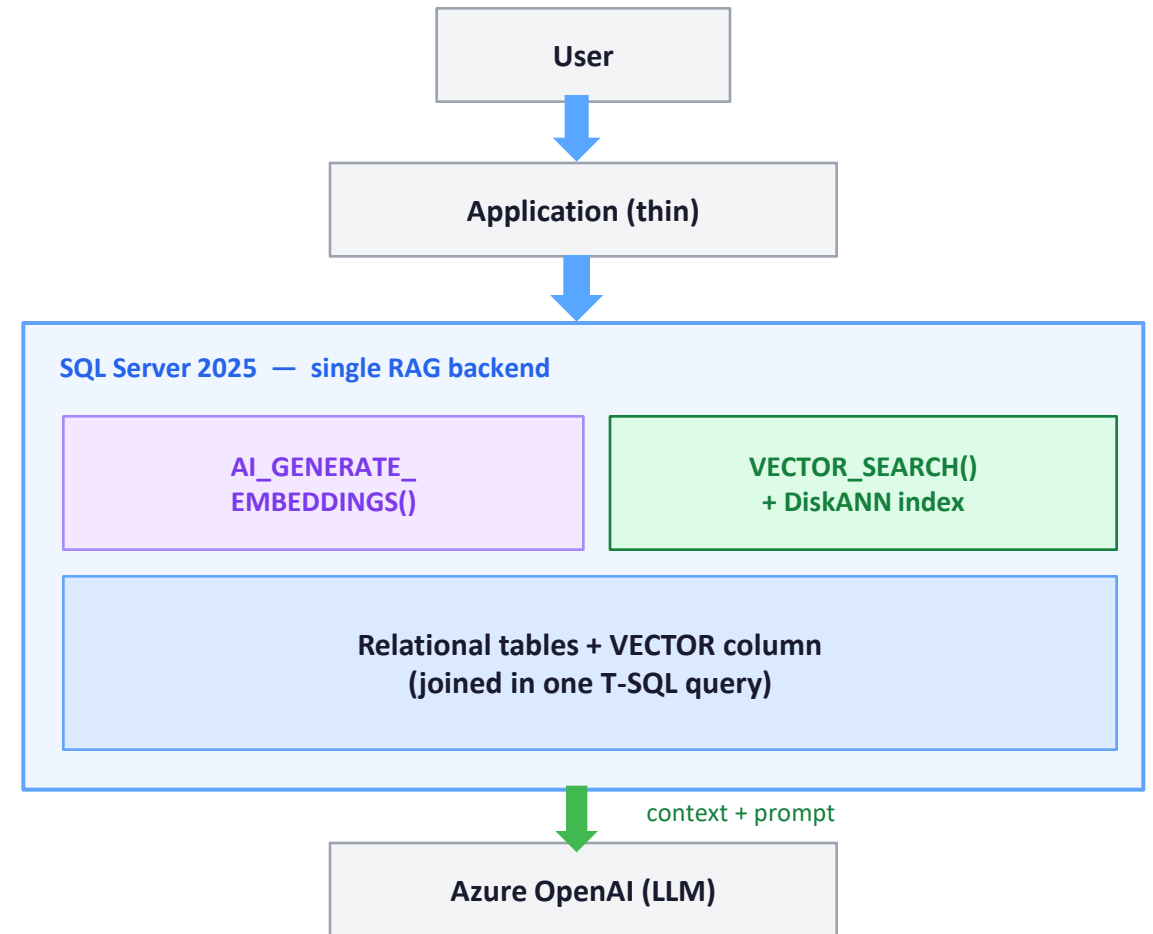
How retrieval-augmented generation hangs together at infrastructure level

✗ Bolted-together RAG stack



- 4 systems, 4 auth boundaries, 4 SLAs
- Embedder + vector store drift from rows
- Orchestrator owns all retrieval logic

✓ SQL 2025 as single RAG backend



✓ One system of record, one security model, no drift

Five new toys, all callable from *Invoke-DbQuery*.

DbQuery

VECTOR(1536)

A native vector column type.

Stores the embedding next to the row it describes. No bolt-on store. Indexable.

EXTERNAL MODEL

A registered handle to a model.

Define once with a name + endpoint + credential. Reference everywhere by everywhere by alias.

AI_GENERATE_EMBEDDINGS

Text in, vector out.

A scalar function. Use it in [UPDATE](#), in [SELECT](#), anywhere.

VECTOR_SEARCH / VECTOR_DISTANCE

Cosine, dot, L2 — pick one.

Set-returning function over a vector column. Sub-second when paired with a paired with a DiskANN index.

CREATE VECTOR INDEX ... DiskANN

An ANN index, in your database.

Approximate nearest neighbour at index speed. Self-join on similarity similarity becomes a slider.

sp_invoke_external_rest_endpoint

HTTPS, from T-SQL.

Chat completions, webhooks, anything. Auth via [DATABASE SCOPED CREDENTIAL](#). Powerful. Suspicious. Both.

• THE PATTERN — POWERSHELL ON BOTH ENDS, AI IN THE MIDDLE

PowerShell's the *hands*. SQL is the *brain*. The *brain*. The model is just a phone call.

01 · INGEST



PowerShell brings the data

Get-Help, Get-DbDatabase, AST walks of your scripts folder. PowerShell already knows how to enumerate everything in your estate.



02 · EMBED & SEARCH



SQL Server thinks about it about it

AI_GENERATE_EMBEDDINGS to vectorise,
VECTOR_SEARCH to query,
sp_invoke_external_rest_endpoint to ask the model
model questions in plain T-SQL.



03 · CONSUME



PowerShell does the next thing

dbatools pulls the answers back, your function decides what to do with them. Same shell you opened five minutes ago.

Three demos in a minute. The pattern is the same one every time — PowerShell in, SQL thinks, thinks, PowerShell out.

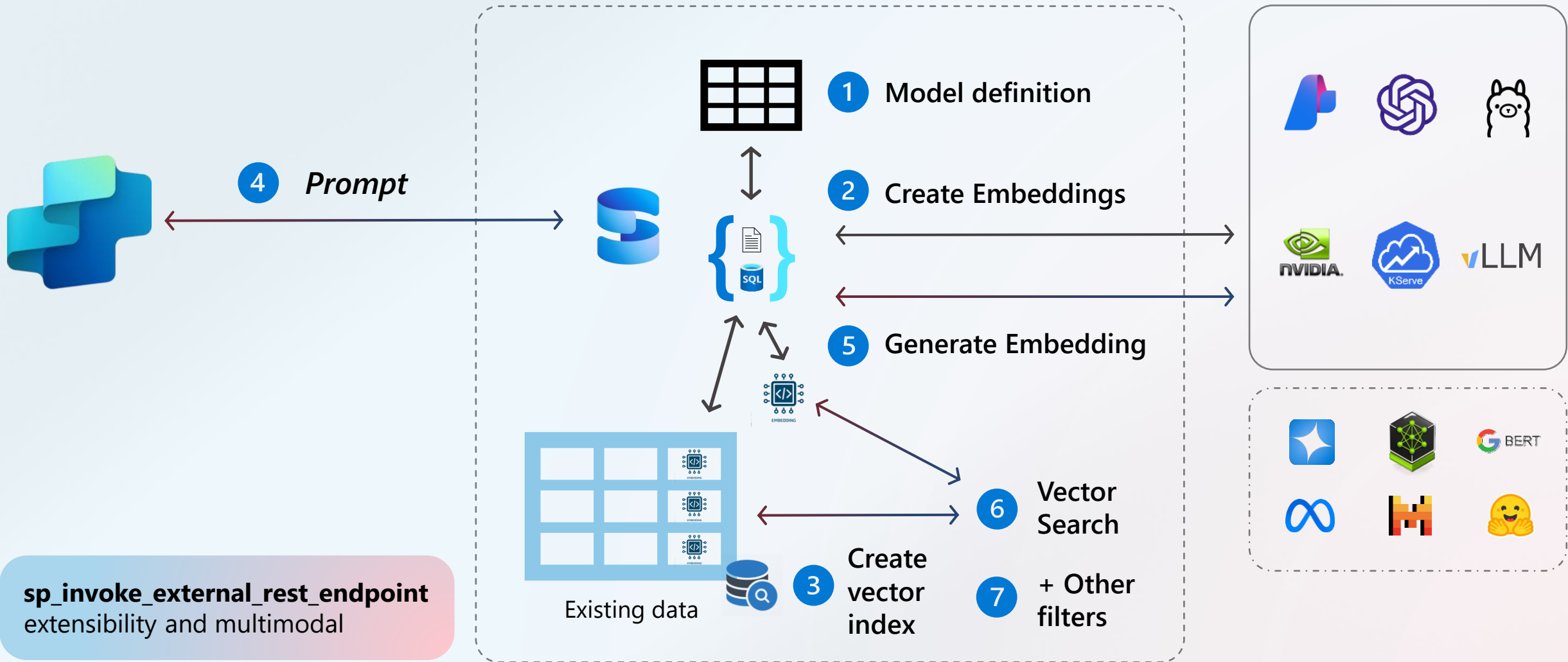
Where PowerShell fits

- **Ingest** — PowerShell pulls the data, SQL embeds and indexes it
- **Search** — PowerShell asks the question, SQL returns the nearest neighbours
- **Orchestrate** — PowerShell composes prompts, SQL stores context, the model responds
- **Govern** — PowerShell does the rounding-up: deployments, permissions, monitoring



SQL Server 2025 Vector Search

API_TYPE
Azure OpenAI
OpenAI
Ollama



Demos

We love Demos!



Q&A

15 minutes



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GIVE ME FEEDBACK



SQL Server 2025, AI and PowerShell Scripts

